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Selectable torque path torque converter

Abstract

An architecture for coupling an output member from a prime mover to an input member of a device to be driven. The architecture includes a torque transfer device defining an indirect coupling for transferring torque therethrough, a direct drive connection member providing a direct coupling for transferring torque therethrough, and a dynamically controllable clutch that selectively and rotationally drives either the torque transfer device or the direct drive connection member. The torque transfer device is rotatably disconnected from the dynamically controllable clutch and undriven when the dynamically controllable clutch is driving the direct drive connection member.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. provisional patent application No. 63/336,946, filed Apr. 29, 2022, the entire contents of which are incorporated by reference herein.

BACKGROUND

1. Field of the Invention

(1) The present invention generally relates to powertrains incorporating a torque converter and, more particularly, to a new architecture of a powertrain system, as might be used in the powertrain of a motor vehicle or another driven device, that incorporates a torque converter. More specifically, the present invention relates to a new architecture of a powertrain system that allows for the selecting of a torque path through the powertrain.

2. Description of Related Art

(2) Today, the automotive industry is increasingly moving away from internal combustion engine (ICE) vehicles and toward electric vehicles. While the technology is improving, one drawback of an all-electric vehicle (EV) is the current limitation on battery technology and, resultantly, the mileage range of such vehicles. Drivers who have mileage needs beyond the range of a typical all-electric vehicle must generally choose between stopping for extended periods of time to recharge the battery or owning a second vehicle, such as an ICE vehicle, that does have an extended driving range.

(3) Hybrid vehicles (HV) offer a blend of ICE vehicle and EV. During use, hybrid vehicles alternate between the ICE and an electric motor to power the vehicle. The net effect of this combined usage is an effective gas mileage that is lower than an EV, but higher than an ICE vehicle. Plug-in, hybrid electric vehicles (PHEV) run on electricity as their primary power source, but will utilize a combustion engine as a backup power source to extend the range of the vehicle.

(4) While the above discussion concerns the powertrains of vehicles, powertrains can also be used in other scenarios where a device requires a rotational input for its operation. One of many other scenarios includes the driving of a gear reduction unit. The output of the gear reduction unit itself may be coupled to other devices also having a need for a rotational input.

(5) A torque converter is a component of the powertrain that couples the prime mover (ICE, electric motor (EM) or ICE/EM combination) of the powertrain with the input of the driven device, e.g. a transmission, a gear reduction mechanism or other device. In a simplistic description, a torque converter uses a hydrodynamic circuit to multiply the input torque and transmit the increased torque as an output torque to the input of the driven device.

(6) The torque converter includes front and rear covers, which together define a shell having an internal chamber where the hydrodynamic circuit is provided. Generally, the output of the prime mover causes rotation of the shell. Rotation of the shell in turn causes rotation of an impeller within the shell. Resultantly, rotation of the impeller directs a hydrodynamic fluid toward a turbine, where the force imparted on the turbine by the fluid rotationally drives the turbine. From the turbine, the fluid is directed radially inward and subsequently axially back toward the impeller. A stator, positioned between the turbine and the impeller, further redirects the fluid so as to efficiently transfer the fluid back to the impeller, thereby multiplying the torque being transferred. Because this circuit is hydrodynamic, a speed differential (known as slip) can occur between the input (impeller) and the output (turbine) sides of the torque converter.

(7) Having both the ability to multiply torque and provide slip is the purpose and function of the torque converter. Torque multiplication is beneficial when launching a vehicle from a dead stop or when towing. However, there are times when it is beneficial to lock-up the torque converter and eliminate slip, a situation where the front cover (the input of the torque converter) is locked with

and directly driving the output of the torque converter. An instance when lock-up is beneficial is when cruising down the road at a steady state. The lock-up condition is achieved by including a lock-up clutch within the torque converter.

(8) During the lock-up condition, all the components of the torque converter rotate together. Torque is transferred through the torque converter not by the hydraulic coupling of the impeller, turbine and stator, but rather by interconnection of the shell/front cover, internal lock-up clutch, output hub and output shaft.

(9) While this lock-up connection allows for better efficiency than the hydrodynamic circuit allows, inefficiencies in the drivetrain exist because the components of the torque converter continue to rotate during lock-up. These inertial losses occur regardless of whether the powertrain incorporates an ICE, EM or ICE/EM as the prime mover. Additionally, the maximum rotational speed of the torque converter is a limiting factor on the rotational speed for operation of the driveline.

SUMMARY

(10) In overcoming the drawbacks and other limitations of the known technologies, in one aspect the present invention provides an architecture whereby the coupling between the output of a prime mover and a driven device is switchable between two different torque flow paths, one including an indirect coupling and one excluding the indirect coupling.

(11) Accordingly, in one aspect the invention provides an architecture for coupling the output member from a prime mover to the input member of a device to be driven thereby.

(12) In another aspect, the architecture includes an input member configured to be connected to the output member of the prime mover; an output member configured to be connected to the input member of the device to be driven; a torque transfer device defining an indirect coupling transferring torque therethrough, the torque transfer device having a device output member connected to the output member of the architecture; a direct drive connection member providing a direct coupling transferring torque therethrough, the direct drive connection being connected to the output member of the architecture; and a dynamically controllable clutch connected to the input member and being selectively and alternately coupled to rotationally drive one of the torque transfer device and the direct drive connection member; whereby when the dynamically controllable clutch is coupled to the torque transfer device, the torque transfer device is rotatably driven by the input member and the output member of the architecture is rotatably driven by the torque transfer device, when the dynamically controllable clutch is coupled to the direct drive connection member the direct drive connection member is driven by the input member, the output member is rotatably driven by the direct drive connection member and the torque transfer device is rotatably disconnected from the input member and undriven.

(13) In a further aspect, the torque transfer device is a torque converter including an impeller and a turbine defining a fluid circuit for driving the turbine by the impeller during operational conditions where torque multiplication and/or slip are desired.

(14) In an additional aspect, the dynamically controllable clutch includes an actuator, a first set of locking elements and a second set of locking elements, In first condition the actuator engages the first set of locking elements with an indirect drive connection member and rotationally driving the torque transfer device. In a second condition, the actuator engages the second set of locking elements and rotationally drives the direct drive connection member. The first set of locking elements are disengaged from the indirect connection member in the second condition, and the second set of locking elements are disengaged from the direct drive connection member in the first condition.

(15) In still another aspect, the first and second sets of locking elements mechanically engage with the indirect drive connection member and the direct drive connection member, respectively.

(16) In yet a further aspect, an electric friction clutch is coupled between the dynamically controllable clutch and an input of the torque transfer device.

(17) In an additional aspect, the electric friction clutch is configured to selectively and frictionally

coupled the torque transfer device to a portion of the dynamically controllable clutch and to cause the torque transfer device to rotate at a speed within a predetermined speed differential relative to a rotational speed of the input member or the direct drive connection member, thereby permitting re-engagement of the torque transfer device and the input member via the dynamically controllable clutch over the operational range.

(18) In yet another aspect, the electric friction clutch includes a first set of friction plates and a second set of friction plates. A translator is configured to compress the first and second sets of friction plates against one another, and an actuator configured to cause the translator to compress the first and second sets of friction plates

(19) In still a further aspect, the one of the first and second sets of friction plates is supported by the dynamically controllable clutch and rotatable therewith, and the other of the first and second sets of friction plates is supported by the indirect drive connection member and rotatable therewith.

(20) In another aspect, the invention provides an architecture having an input member configured to be connected to the output member of the prime mover; an output member configured to be connected to the input member of the device to be driven; a torque converter having an impeller and a turbine defining a fluid coupling for transferring torque therethrough, the torque converter having an output member connected to the output member of the architecture; a direct drive connection member connected to the output member of the architecture and defining a mechanical coupling for transferring torque to the output member of the architecture; and a dynamically controllable clutch connected to the input member and being selectively and alternately coupled to rotationally drive one of the torque converter and the direct drive connection member. When the dynamically controllable clutch is coupled to the torque converter, the torque converter is rotatably driven by the input member and the output member of the architecture is rotatably driven by an output of the torque converter, when the dynamically controllable clutch is coupled to the direct drive connection member the direct drive connection member is driven by the input member, the output member is rotatably driven by the direct drive connection and the torque converter is rotatably disconnected from the input member and undriven.

(21) In yet another aspect, the dynamically controllable clutch includes an actuator, a first set of locking elements and a second set of locking elements. In a first condition the actuator mechanically engages the first set of locking elements with an indirect drive connection member and rotationally drives the torque converter. In a second condition, the actuator mechanically engages the second set of locking elements and rotationally drives the direct drive connection member. The first set of locking elements are disengaged from the indirect connection member in the second condition, and the second set of locking elements are disengaged from the direct drive connection member in the first condition.

(22) In still a further aspect, an electric friction clutch is coupled between the dynamically controllable clutch and an input of the torque converter.

(23) In an additional aspect, the electric friction clutch is configured to selectively and frictionally couple the torque converter to a portion of the dynamically controllable clutch and to cause the torque converter to rotate at a speed within a predetermined speed differential relative to a rotational speed of the input member or the direct drive connection member, thereby permitting re-engagement of the torque converter and the input member via the dynamically controllable clutch over an operational range.

(24) In another aspect, the electric friction clutch includes a first set of friction plates and a second set of friction plates, a translator configured to compress the first and second sets of friction plates against one another, and an actuator configured to cause the translator to compress the first and second sets of friction plates

(25) In yet a further aspect, the one of the first and second sets of friction plates is supported by the dynamically controllable clutch and rotatable therewith, the other of the first and second sets of friction plates is supported by the indirect drive connection member and rotatable therewith.

(26) In another aspect, the architecture, for coupling an output member from a prime mover to an input member of a device to be driven thereby, includes an input member configured to be connected to the output member of the prime mover; an output member configured to be connected to the input member of the device to be driven; a torque converter having an impeller and a turbine defining a fluid coupling for transferring torque therethrough, the torque converter having an output member connected to the output member of the architecture; a direct drive connection member connected to the output member of the architecture and defining a mechanical coupling for transferring torque to the output member of the architecture; a dynamically controllable clutch connected to the input member and being selectively and alternately mechanically coupled to rotationally drive one of the torque converter and the direct drive connection member; an electric friction clutch coupled between the dynamically controllable clutch and an input of the torque converter, the electric friction clutch including a first set of friction plates and a second set of friction plates, a translator configured to compress the first and second sets of friction plates against one another, one of the first and second sets of friction plates being supported by the dynamically controllable clutch and rotatable therewith, the other of the first and second sets of friction plates being supported by the indirect drive connection member and rotatable therewith; and whereby when the dynamically controllable clutch is coupled to the torque converter, the torque converter is rotatably driven by the input member and the output member of the architecture is rotatably driven by an output of the torque converter, when the dynamically controllable clutch is coupled to the direct drive connection member the direct drive connection member is driven by the input member, the output member is rotatably driven by the direct drive connection and the torque converter is rotatably disconnected from the input member and undriven.

(27) In further aspect, the electric friction clutch is configured to selectively and frictionally couple the torque converter to a portion of the dynamically controllable clutch and to cause the torque converter to rotate at a speed within a predetermined speed differential relative to a rotational speed of the input member or the direct drive connection member, thereby permitting re-engagement of the torque converter and the input member via the dynamically controllable clutch over an operational range.

(28) In an additional aspect, the dynamically controllable clutch includes an actuator, a first set of locking elements and a second set of locking elements. In a first condition the actuator mechanically engages the first set of locking elements with an indirect drive connection member and rotationally driving the torque converter. In a second condition, the actuator mechanically engages the second set of locking elements and rotationally drives the direct drive connection member. The first set of locking elements are disengaged from the indirect connection member in the second condition, and the second set of locking elements are disengaged from the direct drive connection member in the first condition.

(29) Further objects, features and advantages of this invention will become readily apparent to persons skilled in the art after review of the following description, including the claims, and with reference to the drawings that are appended to and form a part of this specification.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic illustration of a powertrain system embodying the principles of the present invention.

(2) FIG. 2 is a perspective, cut-away view of a powertrain system embodying the principles of the present invention.

(3) FIG. 3 is cross-sectional view of a powertrain system embodying the principles of the present invention in a state for directly driving the output member of the system.

(4) FIG. 4 is schematic illustration depicting the torque flow path through the powertrain system during the direct drive mode of operation.

(5) FIG. 5 is cross-sectional view of a powertrain system embodying the principles of the present invention in a state for indirectly driving the output member of the system.

(6) FIG. 6 is schematic illustration depicting the torque flow path through the powertrain system during the indirect mode of operation.

DETAILED DESCRIPTION

(7) The new architecture of a powertrain system disclosed herein allows for alternate selection between two torque flow paths: (1) an indirect flow path utilizing a torque converter and (2) a direct-drive flow path. With the direct-drive flow path, the new architecture completely bypasses the torque converter and directly drives the output member of the architecture, which is configured for connecting to the input of a driven device, such as a transmission, a gear reduction unit or other device. In the new architecture, selecting between the two torque paths is achieved using a dynamic controllable clutch (DCC).

(8) When the direct-drive torque path is no longer desired, the new architecture allows for re-connecting and re-establishing of the indirect torque flow path, the flow path through the torque converter. This reconnection may be performed throughout the entire operational speed range of the system. In this regard, an electric friction clutch (EFC) is used to bring the rotational speed of the torque converter to within a specified speed differential with the prime mover (e.g. an electric motor), allowing the dynamic controllable clutch to seamlessly re-establish the indirect torque flow path through the torque converter.

(9) Referring now to the drawings, a new architecture for a powertrain system is shown therein and generally designated at **10**. Torque is provided to the system **10** from the output member **12** a prime mover **14**, such as an internal combustion engine (ICE), electric motor (EM) or other device, to an input member **16** of the system **10**. From the input member **16**, torque is transferred to a selectable switching device **18**, which is a unit incorporating both a dynamic controllable clutch **20** and an electric friction clutch **22**. The switching device **18** in turn causes the torque to be transmitted to an output member **24** of the system **10**, which may be coupled to or provided as the input of device that is to be driven. In transferring the torque, the switching device **18** can transmit the torque to the system's output member **24** indirectly through a torque converter **26** or directly through a mechanical connection that completely by-passes the torque converter **26**. Selection between the two torque paths is achieved using the dynamic controllable clutch **20** of the switching device **18**.

(10) When transferring torque through the torque converter **26** in the indirect mode of operation, torque flows from the dynamic controllable clutch **20** through an "indirect drive" connection element **30** to an input hub **32** of the torque converter **26**. Thereafter, torque is transferred by way of the torque converter's shell **34** to an impeller **36**. The impeller **36** is fluidly coupled with a turbine **38** and a stator **40** to multiply the torque. From the turbine **38**, torque is transmitted to a one-way clutch **42**, which will operate in a lock-up condition to drive the system's output member **24** and the input member (not shown) of the driven device (not shown). The fluid coupling of the torque converter is an indirect coupling in that a rigid, fixed connection is not established thereby.

(11) When transferring torque in the direct mode of operation, torque flows from the dynamic controllable clutch **20** to a "direct drive" connection element **44**, which is coupled to an intermediate output member **46**. The intermediate output member **46** is in turn coupled to the system's output member **24** and the input member of the driven device. In the direct mode of operation, the indirect drive connection element **30** is not driven and the torque converter **26** is completely by-passed. No torque is inputted into the torque converter **26**. The one-way clutch **42** of the torque converter **26**, which is located between the turbine **38** and system's output member **24**, operates in an over-running condition. Accordingly, no components of the torque converter **26** (input hub **32**, shell **34**, the impeller **36**, turbine **38** or stator **40**) are driven or rotate during direct mode operation and no losses occur as a result thereof.

- (12) With existing technology, in order to by-pass the fluid coupling between the impeller and turbine, the output member is mechanically coupled to the rotating shell of the torque converter. This is achieved using a piston/clutch assembly provided internally of the torque converter. When driven in this manner, losses occur since the shell, impeller, turbine and stator all continue to be driven in rotation.
- (13) With the present architecture, the internal piston/clutch assembly (the lockup clutch) of the torque converter is removed and the switching device **18** is added upstream of the torque converter **26**. As a result, the entire torque converter inertia can be avoided in the direct drive mode of operation.
- (14) While discussed herein in combination with a torque converter **26**, the architecture of the present invention may be implemented in applications utilizing torque transfer devices other than torque converters. For example, another device, such as a fluid coupling device, hydrostatic coupling, or other torque transfer devices could be used in place of the torque converter **26**.
- (15) The dynamic controllable clutch **20** operates such that torque imposition results from either an engagement or non-engagement of locking elements **54**, **55** associated with the dynamic controllable clutch **20**. The dynamic controllable clutch **20** includes an actuator **48** that is movable between first and second positions. The actuator **48**, through a spring plate **50** and springs **52**, **52**, acts on the locking elements **54**, **55** to couple the output member **12** from the prime mover **14** to the system output member **24** through either the indirect drive connection element **30** or the direct drive connection element **44**.
- (16) For example, when the actuator **48** is in a first position, locking elements **54** associated with the direct drive connection element **44** are coupled with the input member **16**, and therefore the output member **12** of the prime mover **14**, to the system output member **24** along the torque flow path that is as outlined in FIG. 4. When the actuator **48** is in a second position, the locking elements **55** associated with the indirect drive connection element **30** couple the input member **16** to the torque converter **26** and to the system output member **24** along the torque flow path depicted in FIG. 6.
- (17) The actuator **48** is a linear actuator acting on the spring plate **50**. The springs **52**, **53** engage the respective locking elements **54**, **55** of the direct and indirect drive connection elements **44**, **30** wherein axial movement of the spring plate **50** moves the respective locking elements **54**, **55** between engagement and disengagement with the two connection elements **30**, **44**. The locking elements **54**, **55** associated with the direct drive connection element **44** and the indirect drive connection element **30** mechanically couple the input member **16** to the system output member **24** in at least one direction about the rotational axis.
- (18) Thus, the direct drive connection element **44** includes a first set of locking elements **54** and the indirect drive connection element **30** includes a second set of locking elements **55**, the latter of which are radially spaced outward from the locking elements **54** of the direct drive connection element **44**. As noted above, the second position of the actuator **48** couples the input member **16** to system output member **24** through the torque converter **26** and the first position of the actuator **48** couples the input member **16** to the system output member **24** while bypassing the torque converter **26**. Accordingly, the actuator **48** movement locks the input member **16** to the system output member **24** through the direct drive connection element **44** while simultaneously unlocking the torque converter **26** from the input member **16** through the indirect drive connection element **30** and vice versa.
- (19) While the switching device **18** is directly coupling the input member **16** to the system output member **24**, no rotational input is provided to the torque converter **26**, and the torque converter **26** may cease to rotate. To enable smooth reconnecting of the torque converter **26** with the input member **16**, and the output member **12** of the prime mover **14**, over the operational speed range of the system, the electric friction clutch **22** is used.
- (20) When desiring to switch from the direct drive mode of operation to the indirect drive mode

(while the switching device **18** is still transmitting torque through the direct drive connection element **44** via the dynamic controllable clutch **20**), the electric friction clutch **22** is used to bring the shell **34**, and therefore the impeller **36**, of the torque converter **26** up to a speed that is within a specified speed differential range relative to the rotational speed of the input member **16**. With the shell **34** rotating at a speed in the specified range, the dynamic controllable clutch **20** can smoothly be re-engaged with the indirect drive connection element **30**.

(21) The electric friction clutch **22** is an electromagnetic clutch including a friction pack **56** having friction plates that are used to engage input and output members of the clutch **22**, like a hydraulically actuated, multi-plate, wet friction clutch. The electric friction clutch **22** includes the friction pack **56**, a translator **58** having a translator piston and a translator plate, and a stator **60** having a stator core and a stator coil. The friction pack **56** includes a first set of friction plates **62** that are connected to the translator piston and to an element or bell **64** of the dynamic controllable clutch **20** that is connected to the input member **16** and always rotating. The friction pack **56** also includes a second set of friction plates **66** that are coupled to the indirect drive connection element **30** and input hub **32** of the torque converter **26**. The stator **60** of the electric friction clutch **22** is axially fixed in position and the translator **58** is axially movable. When the translator piston moves toward the friction pack **56** and away from the stator **60**, the friction plates **62**, **66** are compressed together.

(22) Once the friction plates **62**, **66** are compressed, friction between the two sets of friction plates carries torque and, consequently, a power flow path extends through the input **16**, dynamic controllable clutch **20**, the friction pack **56**, the indirect connection **30**, the input hub **32** and the torque converter **26**. The establishment of this power flow path through the friction pack **56** initiates rotation of the torque converter **26** and synchronizes the rotational speed of the torque converter **26** with the rotational speed of input member **16**, all while the system output member **24** is still being driven by the direct drive connection element **44** and shortly before engaging/re-engaging the indirect drive connection element **30** with the dynamically controllable clutch **20**. This power flow path is similar to the torque flow path seen in FIG. **6**, but passes through the friction pack **56** instead of the locking elements **55** associated with the indirect drive connection element **30** and does not drive the input shaft (**6**).

(23) As seen from the above discussion, the dynamic controllable clutch **20** enables selection between multiple torque path and the electric friction clutch **22** allows the torque converter **26** to be synchronized and re-engaged with to the input member **16** throughout the operating speed range of the system.

(24) The illustrated dynamic controllable clutch **20** is a 2-position device and features a one-way clutch in one of its torque path selection positions. In an alternative construction, a dynamic controllable clutch **20** with more than two positions (e.g. a 3-position dynamic controllable clutch) could be used.

(25) The mechanics of the dynamic controllable clutch **20** can allow it to function like a one-way clutch (if the locking elements **54**, **55** are designed to only engage in one direction). The one-way clutch function could be used if desirable (or not at all).

(26) Further, the switching device **18** is shown as a combined system in FIG. **1** of the dynamic controllable clutch **20** and the electric friction clutch **22**. These components of the switching device **18** could, however, be separated into standalone systems, that is, a standalone dynamic controllable clutch system and a standalone electric friction clutch system.

(27) The above description is meant to be illustrative of at least one preferred implementation incorporating the principles of the invention. One skilled in the art will really appreciate that the invention is susceptible to modification, variation and change without departing from the true spirit and fair scope of the invention, as defined in the claims that follow. The terminology used herein is therefore intended to be understood in the nature of words of description and not words of limitation.

Claims

1. An architecture for coupling a prime mover output member from a prime mover to a driven device input member of a device to be driven thereby, the architecture comprising: an architecture input member configured to be connected to the prime mover output member; an architecture output member configured to be connected to the driven device input member; a torque transfer device defining an indirect coupling transferring torque therethrough, the torque transfer device having a torque transfer device output member connected to the architecture output member; a direct drive connection member providing a direct coupling transferring torque therethrough, the direct drive connection member being connected to the architecture output member of the architecture; and a dynamically controllable clutch connected to the architecture input member and being selectively and alternately coupled to rotationally drive one of the torque transfer device and the direct drive connection member; whereby when the dynamically controllable clutch is coupled to the torque transfer device, the torque transfer device is rotatably driven by the architecture input member and the architecture output member is rotatably driven by the torque transfer device, when the dynamically controllable clutch is coupled to the direct drive connection member the direct drive connection member is driven by the architecture input member, the architecture output member is rotatably driven by the direct drive connection member and the torque transfer device is rotatably disconnected from the architecture input member and undriven; and wherein the dynamically controllable clutch includes an actuator, a first set of locking elements and a second set of locking elements, in a first condition the actuator engaging the first set of locking elements with an indirect drive connection member and rotationally driving the torque transfer device, in a second condition the actuator engaging the second set of locking elements and rotationally driving the direct drive connection member, the first set of locking elements being disengaged from the indirect drive connection member in the second condition, the second set of locking elements being disengaged from the direct drive connection member in the first condition.
2. The architecture according to claim 1, wherein the first and second sets of locking elements mechanically engage with the indirect drive connection member and the direct drive connection member, respectively.
3. An architecture for coupling a prime mover output member from a prime mover to a driven device input member of a device to be driven thereby, the architecture comprising: an architecture input member configured to be connected to the prime mover output member; an architecture output member configured to be connected to the driven device input member; a torque transfer device defining an indirect coupling transferring torque therethrough, the torque transfer device having a torque transfer device output member connected to the architecture output member; a direct drive connection member providing a direct coupling transferring torque therethrough, the direct drive connection member being connected to the architecture output member of the architecture; and a dynamically controllable clutch connected to the architecture input member and being selectively and alternately coupled to rotationally drive one of the torque transfer device and the direct drive connection member; whereby when the dynamically controllable clutch is coupled to the torque transfer device, the torque transfer device is rotatably driven by the architecture input member and the architecture output member is rotatably driven by the torque transfer device, when the dynamically controllable clutch is coupled to the direct drive connection member the direct drive connection member is driven by the architecture input member, the architecture output member is rotatably driven by the direct drive connection member and the torque transfer device is rotatably disconnected from the architecture input member and undriven; and an electric friction clutch coupled between the dynamically controllable clutch and a torque transfer input of the torque transfer device.
4. The architecture according to claim 3, wherein the electric friction clutch is configured to

- selectively and frictionally couple the torque transfer device to a portion of the dynamically controllable clutch and to cause the torque transfer device to rotate at a speed within a predetermined speed differential relative to a rotational speed of the architecture input member or the direct drive connection member and permitting re-engagement of the torque transfer device and the architecture input member via the dynamically controllable clutch over an operational range.
5. The architecture according to claim 3, wherein the electric friction clutch includes a first set of friction plates and a second set of friction plates, a translator configured to compress the first and second sets of friction plates against one another and an actuator configured to cause the translator to compress the first and second sets of friction plates.
6. The architecture according to claim 5, wherein one of the first and second sets of friction plates is supported by the dynamically controllable clutch and rotatable therewith, the other of the first and second sets of friction plates is supported by the indirect drive connection member and rotatable therewith.
7. An architecture for coupling a prime mover output member from a prime mover to a driven device input member of a device to be driven thereby, the architecture comprising: an architecture input member configured to be connected to the prime mover output member; an architecture output member configured to be connected to the driven device input member; a torque converter having a rotatable cover enclosing an impeller and a turbine, the impeller and turbine defining a fluid coupling for transferring torque therethrough, the torque converter having a torque converter output member connected to the architecture output member; a direct drive connection member connected to the architecture output member and defining a mechanical coupling for transferring torque to the architecture output member; and a dynamically controllable clutch located exteriorly of the rotatable cover of the torque converter, the dynamically controllable clutch including an actuator, the dynamically controllable clutch being connected to the architecture input member and being selectively and alternately coupled to rotationally drive the torque converter in a first position of the actuator and the direct drive connection member in a second position of the actuator; whereby when the dynamically controllable clutch is coupled to the torque converter, the torque converter is rotatably driven by the architecture input member and the architecture output member is rotatably driven by the torque converter output member, when the dynamically controllable clutch is coupled to the direct drive connection member the direct drive connection member is driven by the architecture input member, the architecture output member is rotatably driven by the direct drive connection member and the torque converter is rotatably disconnected from the architecture input member and undriven.
8. The architecture according to claim 7, wherein the dynamically controllable clutch includes the actuator, a first set of locking elements and a second set of locking elements, in the first position the actuator mechanically engaging the first set of locking elements with an indirect drive connection member and rotationally driving the torque converter, in the second position the actuator mechanically engaging the second set of locking elements and rotationally driving the direct drive connection member, the first set of locking elements being disengaged from the indirect drive connection member in the second condition, the second set of locking elements being disengaged from the direct drive connection member in the first condition.
9. The architecture according to claim 7, further comprising an electric friction clutch coupled between the dynamically controllable clutch and an input member of the torque converter.
10. The architecture according to claim 9, wherein the electric friction clutch is configured to selectively and frictionally couple the torque converter to a portion of the dynamically controllable clutch and to cause the torque converter to rotate at a speed within a predetermined speed differential relative to a rotational speed of the architecture input member or the direct drive connection member and permitting re-engagement of the torque converter and the architecture input member via the dynamically controllable clutch over an operational range.
11. The architecture according to claim 9, wherein the electric friction clutch includes a first set of

friction plates and a second set of friction plates, a translator configured to compress the first and second sets of friction plates against one another and an actuator configured to cause the translator to compress the first and second sets of friction plates.

12. The architecture according to claim 11, wherein one of the first and second sets of friction plates is supported by the dynamically controllable clutch and rotatable therewith, the other of the first and second sets of friction plates is supported by the indirect drive connection member and rotatable therewith.

13. An architecture for coupling a prime mover output member from a prime mover to a driven device input member of a device to be driven thereby, the architecture comprising: an architecture input member configured to be connected to the prime mover output member; an architecture output member configured to be connected to the driven device input member; a torque converter having an impeller and a turbine defining a fluid coupling for transferring torque therethrough, the torque converter having a torque converter output member connected to the architecture output member; a direct drive connection member connected to the architecture output member and defining a mechanical coupling for transferring torque to the architecture output member; and a dynamically controllable clutch connected to the architecture input member and being selectively and alternately mechanically coupled to rotationally drive one of the torque converter and the direct drive connection member; an electric friction clutch coupled between the dynamically controllable clutch and a torque converter input of the torque converter, the electric friction clutch including a first set of friction plates and a second set of friction plates, a translator configured to compress the first and second sets of friction plates against one another, one of the first and second sets of friction plates being supported by the dynamically controllable clutch and rotatable therewith, the other of the first and second sets of friction plates being supported by an indirect drive connection member and rotatable therewith; and whereby when the dynamically controllable clutch is coupled to the torque converter, the torque converter is rotatably driven by the architecture input member and the architecture output member is rotatably driven by the torque converter output member, when the dynamically controllable clutch is coupled to the direct drive connection member the direct drive connection member is driven by the architecture input member, the architecture output member is rotatably driven by the direct drive connection member and the torque converter is rotatably disconnected from the architecture input member and undriven.

14. The architecture according to claim 13, wherein the electric friction clutch is configured to selectively and frictionally couple the torque converter to a portion of the dynamically controllable clutch and to cause the torque converter to rotate at a speed within a predetermined speed differential relative to a rotational speed of the architecture input member or the direct drive connection member and permitting re-engagement of the torque converter and the architecture input member via the dynamically controllable clutch over an operational range.

15. The architecture according to claim 13, wherein the dynamically controllable clutch includes an actuator, a first set of locking elements and a second set of locking elements, in a first condition the actuator mechanically engaging the first set of locking elements with the indirect drive connection member and rotationally driving the torque converter, in a second condition the actuator mechanically engaging the second set of locking elements and rotationally driving the direct drive connection member, the first set of locking elements being disengaged from the indirect drive connection member in the second condition, the second set of locking elements being disengaged from the direct drive connection member in the first condition.
